

KARESUANDO, Sweden

WMO: 020810

Lat: 68.4419N

Lon: 22.4436E

Elev: 1089

StdP: 14.13

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.6	-24.8	-36.1	0.7	-29.9	-29.7	1.1	-24.6	26.3	23.8	21.5	24.8	2.2	320	0.752

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	75.2	59.6	71.0	57.7	67.0	56.0	62.0	70.1	59.8	67.1	57.7	64.7	7.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.0	77.5	63.9	56.6	71.2	62.3	54.6	65.9	60.4	28.3	70.1	26.7	66.7	25.4	64.8	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	16.9	14.1	DB	-36.4	79.3	7.7	3.4	-42.0	81.8	-46.5	83.7	-50.8	85.6	-56.4	88.1	(4)
(5)			WB	-35.1	64.5	7.1	3.0	-40.2	66.6	-44.4	68.3	-48.3	70.0	-53.5	72.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.9	7.1	6.9	15.9	27.1	39.3	50.1	57.0	51.7	43.1	30.2	17.4	11.2		
	DBStd	20.34	15.30	15.55	11.62	8.68	7.62	7.07	6.14	5.88	6.40	8.58	11.99	14.44		
	HDD50	7788	1330	1208	1057	688	344	84	11	50	221	614	977	1204		
	HDD65	12830	1795	1628	1522	1138	797	447	256	414	658	1079	1427	1669		
	CDD50	445	0	0	0	0	13	88	229	102	13	0	0	0		
	CDD65	12	0	0	0	0	0	2	9	1	0	0	0	0		
	CDH74	142	0	0	0	0	9	25	95	13	0	0	0	0		
	CDH80	9	0	0	0	0	0	1	8	0	0	0	0	0		
	WSAvg	6.0	5.3	5.5	6.7	6.2	7.0	7.3	6.1	5.5	6.2	5.6	5.0	5.5		
	PrecAvg	18.0	1.0	0.9	0.9	0.9	1.3	1.9	3.0	2.5	1.7	1.7	1.3	1.0		
	PrecMax	26.2	1.9	2.6	2.4	2.8	3.2	4.5	8.4	6.0	3.7	4.1	3.2	2.8		
(14) Wind	PrecMin	10.6	0.2	0.1	0.1	0.1	0.1	0.4	0.5	0.5	0.3	0.3	0.2	0.2		
	PrecStd	3.4	0.4	0.6	0.5	0.6	0.7	0.9	1.5	1.2	0.9	0.9	0.6	0.6		
	DB	38.7	38.7	41.0	50.5	73.3	77.4	80.6	76.0	63.1	50.0	41.0	39.7			
	MCWB	35.5	34.9	35.9	41.6	55.2	59.0	62.5	60.6	52.6	46.9	39.0	36.1			
	2% DB	33.4	34.5	37.3	45.5	63.5	71.9	76.7	70.8	59.2	47.5	37.0	35.9			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	31.1	31.7	33.5	38.5	50.2	55.9	60.6	58.8	52.1	44.7	35.5	33.5			
	5% DB	30.0	30.5	34.5	42.2	56.5	67.2	72.8	66.2	56.1	44.8	34.4	32.6			
	MCWB	28.5	28.7	31.2	36.6	46.4	53.8	59.2	56.7	50.5	42.3	33.4	31.0			
	10% DB	26.2	27.2	31.5	39.3	51.4	62.6	68.6	62.7	53.4	41.3	32.3	29.9			
	MCWB	25.2	25.8	28.9	34.8	43.5	52.3	57.6	55.4	49.2	39.4	31.7	28.7			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	35.5	35.5	36.2	42.8	55.8	61.8	65.5	64.2	56.1	48.0	39.2	36.5			
	MCDB	38.7	38.4	40.2	49.6	70.5	71.5	75.2	72.0	59.7	49.0	41.1	39.1			
	2% WB	31.5	31.6	33.9	39.2	51.0	58.2	63.0	60.9	53.3	45.1	35.7	33.8			
	MCDB	33.1	34.2	36.9	44.5	61.7	66.6	71.8	66.9	57.3	47.2	36.7	35.7			
	5% WB	28.4	28.8	31.4	37.0	47.2	55.9	60.9	58.2	51.6	42.4	33.5	31.2			
(33) Mean Daily Temperature Range	MCDB	29.9	30.5	34.1	41.4	55.1	64.7	68.9	64.0	55.0	44.5	34.3	32.5			
	10% WB	25.1	25.8	29.1	35.2	44.5	53.4	59.1	56.2	49.8	39.7	31.8	28.7			
	MCDB	26.1	27.2	31.4	39.0	50.7	61.2	66.5	61.9	53.0	41.5	32.4	30.0			
	MDBR	15.5	16.0	18.1	16.8	14.5	15.1	15.8	15.8	13.7	10.5	12.8	14.8			
	5% DB	15.2	14.1	14.7	17.0	23.8	22.7	22.5	22.7	17.9	11.5	9.1	11.9			
(40) Clear-Sky Solar Irradiance	MCWBR	14.0	12.6	11.7	11.2	12.5	10.3	9.1	11.7	11.1	8.8	8.5	10.9			
	MCDBR	14.3	13.7	14.2	15.5	21.6	19.4	19.1	19.3	14.4	10.4	8.7	11.8			
	MCWBR	13.4	12.4	11.6	10.3	11.9	9.9	9.4	11.0	9.6	8.5	8.2	11.1			
	taub	0.180	0.237	0.276	0.305	0.325	0.330	0.336	0.322	0.290	0.254	N/A	N/A			
	taud	1.920	2.096	2.194	2.225	2.405	2.492	2.488	2.509	2.564	2.305	N/A	N/A			
(42) All-Sky Solar Radiation	Ebn at Noon	28	176	240	266	275	277	271	262	245	173	N/A	0			
	Edh at Noon	2	14	24	32	30	29	28	24	17	11	N/A	0			
	RadAvg	9	148	625	1258	1599	1563	1470	1045	561	183	26	0			
(45)	RadStd	1	12	27	56	154	110	142	72	58	28	3	0			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page